EL1101E Nature of Language

AY 25/26 Sem 2 — github/omgeta

1. Introduction

Linguistics is the descriptive scientific study of language as a system and phenomenon using qualitative & quantitative methods. It studies:

i. How language is acquired
ii. How it is processed in the brain
iii. How it can be processed by computers
iv. How it changes over time
v. How it varies by situation
vi. How it functions in society

Big Questions:

i. Do all languages share an underlying structure?
ii. Is the way you think shaped by languages spoken?
iii. Is language learned and processed in the brain differently from other skills?

Knowledge of Language:

i. Function: knowing how to communicate
ii. Form: knowing the words of that language and the rules for putting them together

Key Features of Language:

i. Arbitariness: of form to meaning
ii. Discreteness: of functional units in messages
iii. Rule Governedness: of composition of discrete units
iv. Productivity: using finite grammar to compose infinite number of utterances

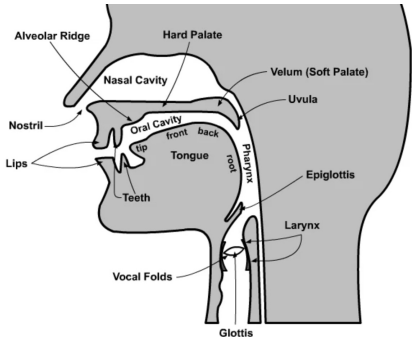
2. Phonetics

Phonetics is the study of speech sounds. Areas:

i. Articulatory: how speech sounds are produced
ii. Acoustic: acoustic properties of the speech signal
iii. Auditory: listeners' perception of speech sounds

Speech Production:

i. Voice is powered by air from the lungs
ii. Speech is the molding of air by the vocal tract as it travels to escape through our mouth and nose
iii. Voiced/Voiceless: vocal folds vibrate/ don't vibrate
iv. Nasal/Oral: velum lowered/ raised



IPA (International Phonetic Alphabet):

i. Symbols representing speech sounds, defined by articulatory production
ii. Universal one-to-one correspondence to sounds

Consonants: airflow via oral cavity is obstructed

i. Voicing (voiced vs. voiceless)
ii. Place of articulation (location of obstruction)
iii. Manner of articulation (cause of obstruction)

Vowels: airflow via oral cavity is unobstructed

i. Rounding: rounded vs. unrounded lips
ii. Height: tongue height (high, mid, low)
iii. Backness: tongue position (front, central, back)
iv. Tenseness: effort (tense, lax)
v. Monophthongs vs. diphthongs vs. diphtongized vowels (steady, transient, micro-transient optional)

Places of Articulation (English):

i. Bilabial: obstruction made with two lips
ii. Labiodental: lower lip against upper teeth
iii. Interdental: tongue between teeth
iv. Alveolar: tongue at/near alveolar ridge
v. Post-alveolar: tongue just behind alveolar ridge
vi. Palatal: tongue body against/near hard palate
vii. Velar: tongue body against/near velum
viii. Glottal: constriction at glottis

Manners of Articulation (English):

i. Stops: complete closure of oral cavity
   - Plosive: followed by release of closure
   - Nasal: airflow redirected through the nose
ii. Fricatives: narrow constriction causing turbulence
iii. Affricates: plosive → fricative as a single sound
iv. Approximants: narrowing but no turbulence
   - Central: air flows through center of mouth
   - Lateral: air flows through sides, as tongue tip touches alveolar ridge
v. Flap: tongue briefly touches roof of mouth

3. Phonology

Phonology is the study of the structures and patterns of speech sounds within languages.

Representations:

i. Phoneme: abstract distinctive unit in a language; replacement changes meaning (e.g. /t/)
ii. Phone: actual speech sound
   - Allophone: variant realization of a phoneme; does not change meaning (e.g. [t], [ɾ], [t^h])
iii. Language-specific: allophones vary

Phone Distributions:

i. Contrastive: two phones can occur in the same phonological environment ⇒ different phonemes
   - Minimal pairs: words differ by only one sound
ii. Complementary: two phones never occur in the same phonological environment ⇒ likely allophones
   - Must also be phonetically similar (share features)
iii. Free: non-contrastive but can occur in same phonological environment ⇒ still allophones
   - Not determined by the phonetic environment but by non-phonetic factors (e.g. formality)

Phonemic Analysis (assuming no free variation):

i. Look for minimal pairs
   - If found: the two sounds are contrastive ⇒ allophones of different phonemes
ii. Examine phonetic environments: for each sound, list environments of the form X__Y
   - Can you state a phonetic generalisation over where each sound occurs?
   - If yes: sounds are in complementary distribution ⇒ likely allophones of the same phoneme
   - If no: no evidence for allophony ⇒ allophones of distinct phonemes
   - Generalisations can be sensitive to any phonetic parameter: e.g. voicing, place
iii. Identify the phoneme: conventionally the less restricted allophone represents the phoneme

Phonotactics are language-specific constraints on arrangement of phones:

i. Constraints on possible sound sequence
   - e.g. English [g] cannot precede [n]
ii. Constraints on sounds in particular positions
   - e.g. [ŋ], [ʒ] cannot occur syllable-initial
iii. Constraints on possible syllable types
   - e.g. Hawaiian only allows V and CV syllables
iv. English allows 0–3 consonants at the start or end of a syllable, but not any sequence is permitted
v. Allophony is a reflex of phonotactics – the distribution of allophones is governed by rules on how a phoneme is produced in different positions
   - e.g. English /k/ → [k^h] syllable-initial else [k]

Impact on Language Learning:

i. Different sound inventories: speakers may substitute an unfamiliar sound with a similar one from their native language
   - e.g. [ɲ] → [n]
ii. Different phonotactic constraints: speakers may (unknowingly) impose the phonotactic constraints of their native language onto a second language
   - e.g. English speakers aspirating word-initial /p/ as [p^h] in Malay

4. Morphology

Morphology is the study of words and their structure.

Morpheme is the smallest meaningful unit in a language:

i. Same form ≠ same morpheme if meanings differ
   - e.g. un_1 ('to reverse an action') in *undo, untied* vs. un_2 ('not') in *unfair, unsafe*
ii. Free morpheme: can occur alone as a word
   - e.g. *do, tie, fair, safe*
iii. Bound morpheme: cannot occur alone on its own
   - e.g. *un-, -ed, -s*
iv. Simple words containly only one morpheme, complex words contain multiple

Allomorphs:

i. Slightly different forms of the same morpheme – same meaning, phonetically similar and are in complementary distribution with one another
   - e.g. indefinite article: [ə] before consonants, [ˈæn] before vowels
   - e.g. past tense: [t] (*plucked*), [d] (*hugged*), [əd] (*rated*)

Inflectional vs. Derivational Morphemes:

i. Inflectional: convey grammatical information; only 7-8 inflectional affixes in English:
   - 3rd person singular present tense (*he walk-s*)
   - progressive (*walk-ing*)
   - past participle (*has walk-ed*)
   - past tense (*walk-ed*)
   - plural (*cat-s*)
   - comparative (*tall-er*)
   - superlative (*tall-est*)
   - possessive (*John's*)
ii. Derivational: create a new word, often with a different meaning and/or word class
iii. Rule: derivation occurs closer to the root than inflection (e.g. *piglet-s* not **pigs-let*)

Roots, Stems, Compounds:

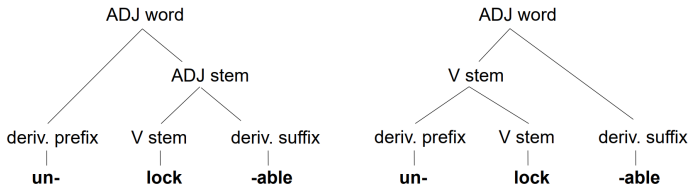
i. Root: primary indivisble morpheme
   - In English, almost always free morphemes
ii. Stem: unit that an affix attaches onto
   - Can be root alone, or root + affix(es)
iii. Compound Word: multiple stems combined
   - Orthography is irrelevant (e.g. hyphens)

Affixes - bound morphemes that attach to stems:

i. Prefix: occurs before the stem (e.g. *un-tie*)
ii. Suffix: occurs after the stem (e.g. *tie-d*)
iii. Infix: occurs inside the stem
iv. Circumfix: occurs around the stem (prefix + suffix as a single morpheme)

Word Structure:

i. Affixes have constraints on what classes they attach to and what they create
ii. Multiple affixation can force an order of formation (intermediate forms must be well-formed)
iii. Word structure is hierarchical:
   - Binary-branching
   - Each node has a label
   - Each stage must be a well-formed word
iv. Structural Ambiguity: multiple possible internal structure, each associated with a different meaning (e.g. *unlockable*: 'able to be unlocked' vs. 'not able to be locked')



Word Formation:

i. Affixation
ii. Compounding: two or more free stems
   - e.g. *salesman, manspread*
iii. Conversion: change word class with no changes
   - e.g. *Google* (N)→*google* (V)
iv. Clipping: removing part of a word
   - e.g. *sister* → *sis*, *McDonald's* → *Mac's*
v. Back-formation: removing what looks like (but wasn't) an affix; special case of clipping
   - e.g. *editor* > *edit*
vi. Blending: combining a clipping with another clipping or stem
   - e.g. *bro(ther)* + *romance* → *bromance*; *man* + *explaining* → *mansplaining*
vii. Acronym: initials pronounced as a word
   - e.g. LASER, [lɔl] lol
viii. Initialism: initials pronounced as individual letters
   - e.g. MRT, NUS, KFC
ix. Eponyms: formed from people's or trade names
   - e.g. *sadism, Google, Kleenex*
x. Borrowing: taken from another language, often with adapted meaning or pronunciation
   - e.g. *manga, taco, anime*
xi. Reduplication: repeating part or all of a word
   - e.g. Malay *langit* 'sky' → *langit langit* 'ceiling'

5. Word Classes

Word Classes categorise words based on grammatical behaviour, with sufficiency tests:

i. Noun: entities
   - Morphological: can be plural/ possessive marker
   - Distributional: can fill "a __ of __"; can be preceded by determiners and adjectives in a NP
ii. Verb: actions
   - Morphological: can be marked for tense; can be marked for progressive aspect; exhibits 3rd-person singular agreement (-s)
   - Distributional: appears after subject; can negate with *do not*; can appear after auxiliary verbs
iii. Adjective: modify nouns
   - Morphological: can be comparative/ superlative
   - Distributional: can occur between a determiner and a noun; can follow *more/most*; can co-occur with other adjectives
iv. Adverb: modify verbs, adjectives and adverbs
   - Morphological: can be comparative/ superlative
   - Distributional: freedom of positioning
v. Preposition: indicate time/place/direction/manner
   - Distributional: precedes a pronoun or NP; does not form an NP with the following noun
vi. Determiner: specify nouns
   - Distributional: obligatorily precedes a noun (and optionally adjectives) within a NP; cannot appear alone without a following noun
vii. Pronoun: stand in for noun phrases
viii. Conjunction: link words/phrases/clauses

Content vs. Function Words:

i. Content (nouns, verbs, adj., adv.):
   - Open-class: can easily add new words
   - Contentful: easily defined meaning
ii. Function Words (det., pronouns, conj., prep.):
   - Closed-class: lexicon relatively stable
   - Serve grammatical functions

6. Syntax

Syntax is the study of rules underlying sentence structure.

i. Grammaticality does not depend on acceptability/meaningfulness of the sentence

Constituents are chunks that operate as a single unit:

i. Types: words, phrases, sentences
ii. Constituency Tests:
   - Q&A: if string can answer a question
     - *What did Kim buy? that book*
   - Topicalization: if a string can appear upfront
     - *That book, Kim bought with her first wages*
   - Clefts: *It is X that Y*
     - *It was that book that Kim bought with her first wages*
   - Coordination: if a string can be coordinated with another string from the same phrasal category
     - *Kim bought that book and that pen*
   - Replacement: replace chunk with a pronoun/ known constituent (e.g. *it*, *do so*)
     - *Kim bought it*
   - – Not 100% reliable, but passing > 1 test is strong evidence for constituency

Phrase is an intermediate between words and sentences:

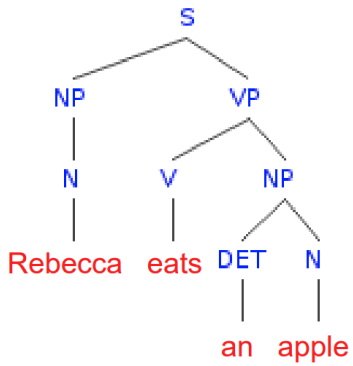
i. Head: determines the phrase's syntactic class:
   - Noun Phrase (NP): replaceable by a pronoun
   - Verb Phrase (VP): replaceable by *slept* (meaning changes) or *do/did so* (meaning preserved)
   - Prepositional Phrase (PP): P + NP sequence; can modify VP or N within NP
ii. Head Test: "A Y is a type of X"
   - "The yellow box" is a type of "box"
   - "ate an apple" is a type of eating
   - "in the little box" is a type of in-position

Sentence Constituents:

i. Subject (NP)
ii. Predicate (VP)
   - Argument: necessary for predicates' meaning
   - Complement: non-subject argument
   - Types:
     - Intransitive (IV): subject only
     - Transitive (TV): subject + complement
     - Ditransitive (DV): subject + two complements
     - Sentential Compl. (SV): subject + sentence
   - Adjunct: optional; adds information about how something occurred

Phrase Structure:

i. Rules indicate what types of constituents can make up different classes of phrases. In English:
   - S → NP VP (Complete Sentence)
   - VP → SV S
   - VP → V (NP) (NP) (PP)
   - NP → (Det) (Adj)* N (PP)
   - PP → P NP
   - X → X Conj X
ii. Syntax Trees: represent hierarchical structure
   - A string is a constituent ⇔ there is a node that exclusively dominates that sequence



Structural Ambiguity: arises from different possible structures

i. Commonly due to PP attachment (to VP or NP)
ii. [VP saw [NP the man] [PP with the telescope]] - used a telescope to see
iii. [VP saw [NP the man [PP with the telescope]]] - saw a man carrying a telescope

Lexical Ambiguity: arises from word with multiple meanings

i. *Molly owns a black bat* — *bat* is ambiguous, but *a black bat* remains an NP

7. Semantics

Semantics is the study of meaning in language.

Word Meaning Relationships:

i. Similar Form, Different Meanings:
   - Homophone: same pronunciation, diff. spelling
   - Homograph: same spelling, diff. pronunciation
   - Homonym: same spelling and pronunciation
   - Polyseme: same word used in diff. related sense

ii. Hyponomy: if $A \subseteq B$
   - A is a hyponym of B
   - B is a hypernym of A

iii. Meronymy: part-whole relationship
   - A is a meronym of B (A is a part of B)
   - B is a holonym of A (B is the entire unit)
   - e.g. *finger* is a meronym of *hand*; can be abstract

iv. Similarity/Opposition:
   - Synonyms: words with the same/similar meaning
   - Antonyms: words with opposite meaning
     * Gradable: ends of a continuous scale
     * Complementary: no middle ground
     * Reverses: opposite roles in a relationship
     * Converses: two perspectives on a relationship

Sentence Meaning Relationships:

i. Entailment: if A is true, B must be true
   - *Taylor bought a poodle* $\Rightarrow$ *Taylor bought a dog*

ii. Contradiction: if A is true, B must be false
   - *Taylor bought a dog* $\Rightarrow\Leftarrow$ *Taylor didn't buy a dog*

iii. Paraphrase: A and B entail each other
   - *He killed it* $\Leftrightarrow$ *It was killed by him*

iv. Presupposition: If A presupposes B, then both A and *not* A still assume B
(background assumption that survives negation)
   - *Lee is (not) the current Prime Minister of Singapore* $\Rightarrow$ *Singapore has a Prime Minister*

Layers of Meaning:

i. Denotation vs. Connotation
   - Denotation: the objective thing referred to
   - Connotation: associations evoked by a word
   - e.g. *home* vs. *residence*: same denotation, different connotations
   - e.g. *pro-life* vs. *anti-abortion*: same denotation, different (positive vs. negative) connotations

ii. Extension/Reference vs. Intension/Sense
   - Extension/Reference: the object/entity in the world an expression points to
   - Intension/Sense: the intrinsic meaning of an expression
   - Same extension, different intension: *Arthur the Red Panda* and *the official mascot of FASS* refer to the same entity but have different intrinsic meanings
   - Non-referring expressions have intension but no extension
     (e.g. *dragon*, *phoenix*)

iii. Shifting Extension: fixed intension, but extension varies by context/speaker
   - Personal pronouns: *she* always means 3rd person singular female (fixed intension), but refers to different individuals across utterances
   - Demonstratives: *here/there*, *this/that* — fixed sense (near/far from speaker), shifting reference

8. Pragmatics

Pragmatics is the study of meaning in context.

Speech Acts:

i. Utterances convey meaning and make listeners do specific things
   - Not all utterances have truth values
   - Saying something is doing something
   - Different words can achieve the same outcome
   - Utterances can be the same but with different intent or effect depending on context

ii. Components:
   - Locution: utterance
   - Illocution: intention
   - Perlocution: effect

iii. Performatives: not true/false; change social reality under particular (felicity) conditions
   - *I now pronounce you man and wife.*
   - "Hereby" test: *I hereby apologize* vs. **I hereby bought a phone*

iv. Searle's Speech Act Classification:
   - Representative: express beliefs (true/false)
   - Directive: get the addressee to do something (incl. questions as a subtype)
   - Commissive: commit speaker to future action
   - Expressive: express emotional state
   - Declaration: bring about a state of affairs

Conversational Implicature:

i. Cooperative Principle: safe assumption that interlocutors contribute appropriately to the accepted purpose/direction of the exchange

ii. Gricean Maxims:
   - Quality: be truthful
   - Quantity: be as informative as required
   - Relation: be relevant
   - Manner: be clear and orderly

iii. Flouting Maxims:
   - Generates implications via cooperative principle
     * Context-dependent
     * Cancellable
   - Quantity (under-informative):
     * A: *Did you contact the prof and the tutor?*
       B: *I contacted the tutor.*
     * Implicature: B did not contact the prof
   - Quality (sarcasm):
     * A: *Do I look okay in this dress?*
       B: *No, it's so hideous.*
     * Implicature: B thinks A knows perfectly she looks great (B's true views are obvious to A)
   - Relation (surface irrelevance):
     * A: *Can we meet up on Friday?*
       B: *My sister wants me to go shopping with her.*
     * Implicature: B can't meet up on Friday
   - Manner (obscuring / lack of clarity):
     * Journalist: *...what have you got wrong, so that you get it right next time?*
       Boris: *I think, Laura, when you look back...*
     * Implicature: the speaker does not wish to cooperate with the narrative implied

iv. Tests:
   - "and not" test: implicatures are cancellable
     * A: *My sister has been to Taiwan, and I have too.*
   - "but" test: can make the implicature explicit
     * *My sister has been to Taiwan, but I haven't.*

Politeness Theory:

i. Politeness: interacting harmoniously; avoiding conflict/offense (culture- and context-dependent)

ii. Politeness Strategies:
   - Positive Politeness: show friendliness/closeness
   - Negative Politeness: avoid offense with deference

iii. Face Wants:
   - Positive Face: want to be liked/approved of
   - Negative Face: want freedom of action; freedom from imposition

iv. Face-Threatening Acts (FTAs)
   - Threaten addressee: criticism (positive face), orders (negative face)
   - Threaten speaker: apologizing (positive face), thanking (negative face)

v. Mitigating FTAs (examples):
   - No mitigation: *You've got dirt on your nose.*
   - Positive politeness: *This is really interesting, but I have to get going.*
   - Negative politeness: *I'm so sorry to interrupt, but I have to get going.*
   - Off-record hint: *Wow, look at the time.*

vi. Strategy Choice Factors:
   - Social distance
   - Power difference
   - Cost of imposition

9. Sociolinguistics

Sociolinguistics is the study of how language functions in society.

Variation in Language:

i. Accent: difference in pronunciation
ii. Dialect: difference in pronunciation-related features, lexicon and grammatical structure
iii. Mutual Intelligibility:
    - Not straightforward: can be asymmetric; shaped by experience and attitudes
    - Sociopolitical factors often determine whether varieties are treated as "languages" vs. "dialects"
    - "A language is a dialect with an army and navy."
iv. Sociolinguistic Knowledge: native speakers know how to use language appropriately in different situations
v. Language Use Factors:
    - Linguistic: feature environment
    - Social: difference in speakers
    - Stylistic: choice across situations

Non-standard Language:

i. Language Attitudes:
    - Status: standard speakers rated "intelligent/wealthy/educated"
    - Solidarity: non-standard speakers rated "friendly/trustworthy/kind"
ii. Regional features can index local identity and become commodified
iii. Standard varieties are not inherently better:
    - Misconception: non-standard language is "random/illogical/simplified"
    - All varieties are equally rule-governed, expressive
    - Complexity is not owned by the standard
    - Stigma is social: prestige tracks the social status of the people who use a variant

Identity and Style:

i. Co-constitutive: language reflects identity *and* language helps construct identity
ii. Speaker-focused approach: speakers actively construct styles using linguistic + other resources
iii. Personae: recognizable community identities with conventional attributes
    - Ex. SG: Ah Beng, Taxi Uncle, XMM, Auntie
iv. Indexing and style-building:
    - Features become linked with social meanings; a single feature can index many meanings
    - People mix-and-match features to create new styles

Language Change:

i. Diachronic Linguistics: language change across time
ii. Synchronic Linguistics: language at a time-slice
iii. Languages change incrementally; studied via:
    - Real-time: compare data across time periods
    - Apparent-time: compare age groups at one time

Language Contact:

i. Borrowing: lexical/structural influence
ii. Code-switching: structured alternation in discourse
iii. Contact Languages:
    - Pidgin: reduced system for communication between groups without shared language; no native speakers
    - Creole: stable language that develops from a pidgin and becomes a native language

Language and Justice:

i. Linguistics can be applied to injustice (e.g. forensic linguistics; discrimination; institutional bias)
ii. Accent-based discrimination: differential treatment by accent (e.g. housing enquiries)
iii. Language and police bias: body-cam analysis used to study differences in officer speech

10. Psycholinguistics

Psycholinguistics is the study of the psychology of language.

Language Acquisition:

i. Studying child language is hard ⇒ specialized methods (e.g. head-turn preference)

ii. Early perception:
- Newborns: familiar with rhythms/sounds of caregiver language(s) (womb exposure)
- Perceptual narrowing: perception becomes specialized to native phonology
  - 6 mo: can hear non-native contrasts (e.g. /r/ vs. /l/ for Japanese)
  - 12 mo: reduced sensitivity to non-contrastive differences

iii. Milestones (typical):
- 6 mo: babbling; responds to name
- 12 mo: one-word/holophrastic; understands simple instructions
- 18–24 mo: rapid vocab growth; two-word combinations
- 3–5 yrs: multiword sentences; large receptive/expressive vocab; all vowels + most consonants
- 12 yrs: receptive ∼20k–50k

iv. Learning problems/strategies:
- Mapping problem: many words/day; few directly taught
- Whole object assumption ("gavagai")
- Comprehension > production: *fis* phenomenon

Acquisition Evidence:

i. Error patterns:
- Overregularization: irregulars → rule over-application (go → *goed*; mouse → *mouses*)
- Suggests active rule-building (Go + *-ed* = *goed*)

ii. Universal Grammar + poverty of the stimulus:
- Claim: input lacks enough evidence
- Example:
  - *Anyone who is interested can see me later.*
  - *Can anyone who is interested see me later?*
  - **Is anyone who interested can see me later?*

iii. A critical period?
- Idea: sensitive window (duck imprinting)
- Evidence: age-of-arrival vs. grammar performance (e.g. Johnson & Newport 1989)
- Critics: gradual decline; children still take years; adults have less time

Language in the Brain:

i. Methods:
- Atypical/impaired brains (e.g. stroke)
- Typical brains (direct imaging, e.g. fMRI)
- Indirect evidence from cognition (e.g. speech errors: *"Three cheers for our queer old dean!"*)

ii. Localization (left hemisphere):
- Broca's area: important for speech production
- Broca's aphasia: comprehension relatively intact; difficulty producing speech/syntax
- Wernicke's area: associated with comprehension
- Wernicke's aphasia: fluent syntax but often nonsensical; frequent word substitutions

iii. Beyond Broca/Wernicke:
- Language mostly left-lateralized, but right hemisphere crucial for some tasks (e.g. intonation/emotion)
- Reading Chinese characters involves both hemispheres more than alphabetic reading
- Many additional areas contribute (e.g. switching between languages)

Language Production and Processing:

i. Automatic processing can interfere with responses:
- Stroop effect: word meaning interferes with naming ink color

ii. Multimodal integration:
- McGurk effect: audio [ba] + visual [ga] ⇒ many hear [da]

iii. Modeling Production with connectionist networks
- Nodes + spreading activation
- Most activated node is selected as output
- Ex.: semantic features of *cat* → lexical CAT (vs. DOG) → phonological /k/ /a/ /t/

Language and Thought:

i. Competing views:
- Linguistic determinism (strong Sapir-Whorf): can only conceive what language can describe
- Linguistic relativism (weak Sapir-Whorf): some cognition is influenced by language
- Linguistic universalism: thinking is not affected by language differences (universal metalanguage)

ii. Thinking for speaking: speakers attend to distinctions their language requires
- Color terms:
  - Japanese: one term can cover "blue"/"green"
  - Russian: green vs. light blue vs. dark blue

iii. Metaphorical Mappings:
- Space → time mappings vary; influence cognition
- Horizontal Metaphors (English/Chinese): *looking back/behind* vs. *looking forward/ahead*
- Vertical Metaphors (Chinese): (up month = last month), (down month = next month)
- Boroditsky (2001): English faster after horizontal tasks; Chinese faster after horizontal *or* vertical tasks
- Trainability: English speakers trained with "Monday is above Tuesday" shift toward the Mandarin-like pattern